

LAB # 04

Clauses

Objective:

To understand and implement different SQL clauses (WHERE, ORDER BY, GROUP BY, HAVING, FROM and LIMIT) using WampServer and phpMyAdmin.

Introduction:

In Structured Query Language (SQL), clauses are special components of a statement. They define conditions, limits, sorting, grouping and other rules that are applied to a query, so that only the required data is returned in the required form. This lab explores six commonly used clauses:

- **WHERE** : filters rows based on a specified condition.
- **ORDER BY** : sorts the result set in ascending or descending order.
- **GROUP BY** : groups rows that share a value, often used with aggregate functions such as COUNT().
- **HAVING** : filters the groups created by GROUP BY, just as WHERE filters individual rows.
- **FROM** : specifies the table (or tables) from which data is retrieved.
- **LIMIT**: restricts the number of rows returned by a query, optionally with an offset.

Procedure:

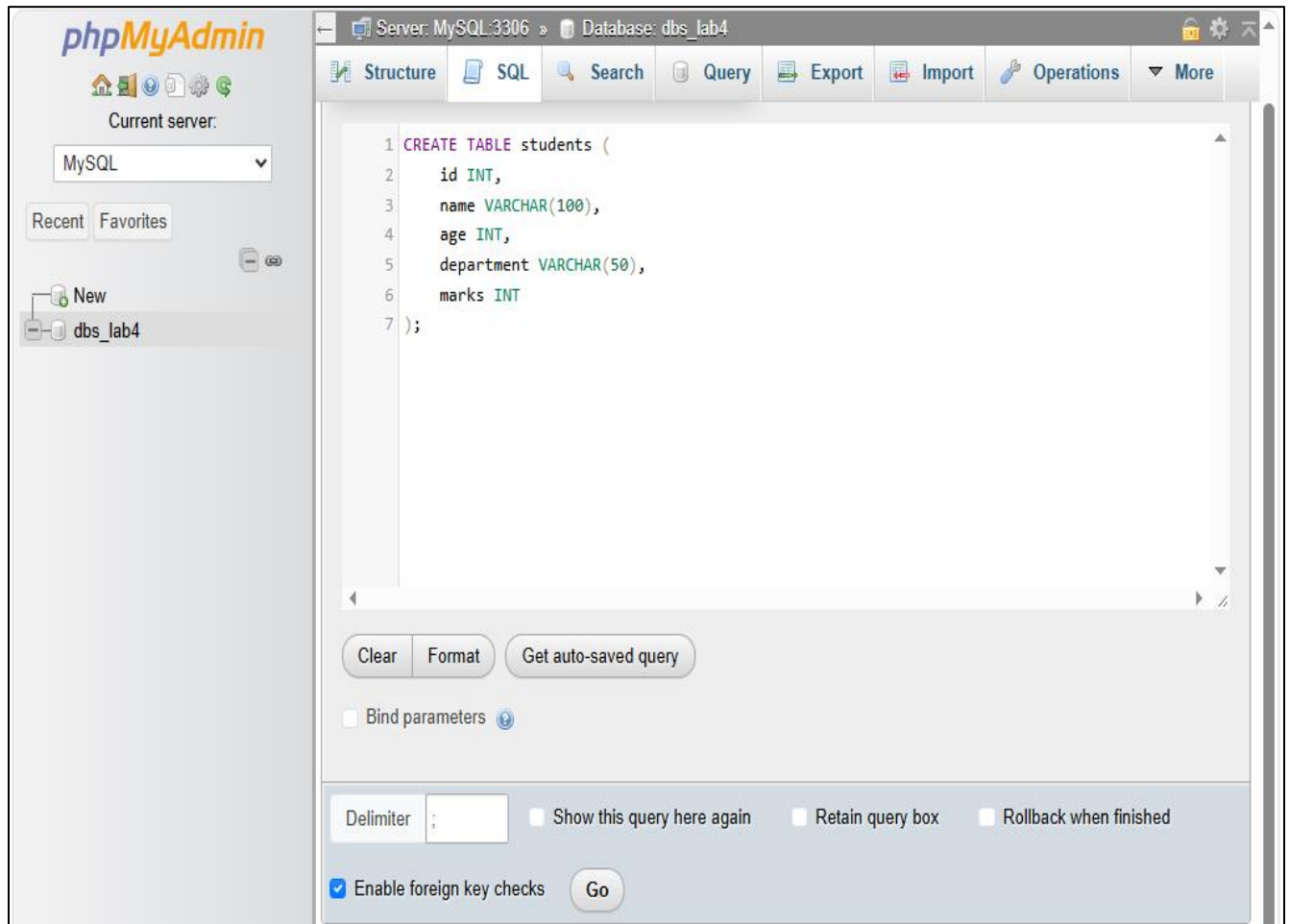
1. Start WampServer and open phpMyAdmin in the browser (localhost/phpmyadmin).
2. Create the database dbs_lab4 and select it.
3. Open the SQL tab and create the students table with the columns id, name, age, department and marks.
4. Insert nine student records into the table using INSERT INTO.
5. Run each clause query (WHERE, ORDER BY, GROUP BY, HAVING, FROM and LIMIT) one by one in the SQL tab.
6. Take a screenshot of each query and of its output.

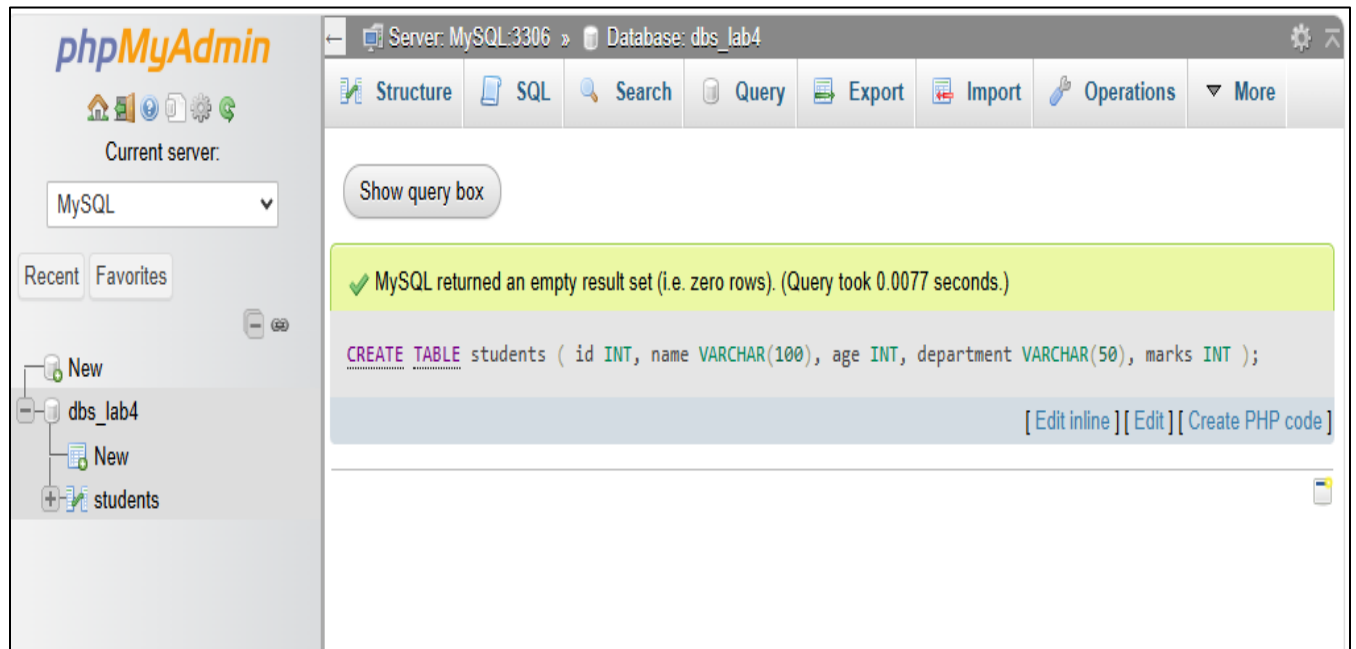
LAB TASK

Step 1: Create Database and Table

A database named "dbs_lab4" was created, and a table named "students" was created inside it with five columns: id, name, age, department and marks.

CREATE TABLE students (id INT, name VARCHAR(100), age INT, department VARCHAR(50), marks INT);





Output: Table created successfully

Step 2: Insert Records

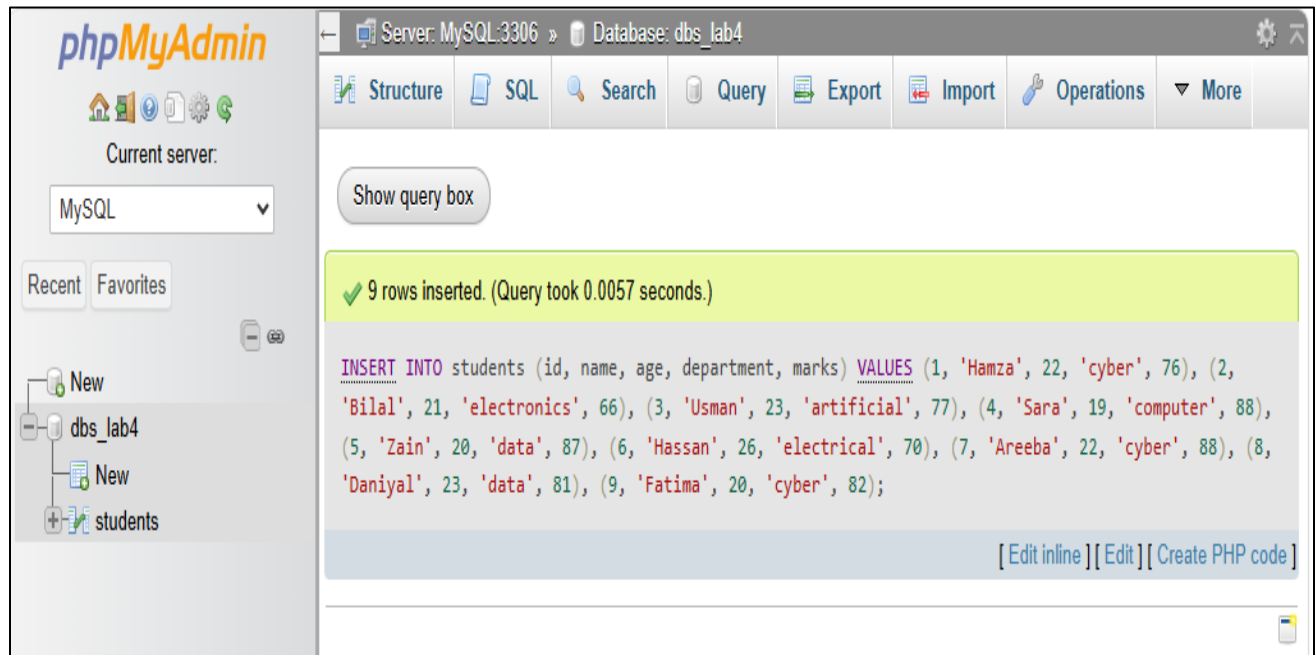
Nine records were added into the "students" table using the INSERT INTO statement.

Insert query: INSERT INTO students (id, name, age, department, marks) VALUES (1, 'Hamza', 22, 'cyber', 76), (2, 'Bilal', 21, 'electronics', 66), (3, 'Usman', 23, 'artificial', 77), (4, 'Sara', 19, 'computer', 88), (5, 'Zain', 20, 'data', 87), (6, 'Hassan', 26, 'electrical', 70), (7, 'Areeba', 22, 'cyber', 88), (8, 'Daniyal', 23, 'data', 81), (9, 'Fatima', 20, 'cyber', 82);

The screenshot shows the phpMyAdmin web interface. The top navigation bar includes links for Browse, Structure, SQL, Search, Insert, Export, Import, and More. The current view is the SQL editor for the 'students' table in the 'dbs_lab4' database. The SQL query being entered is:

```
1 INSERT INTO students (id, name, age, department, marks)
  VALUES
2 (1, 'Hamza', 22, 'cyber', 76),
3 (2, 'Bilal', 21, 'electronics', 66),
4 (3, 'Usman', 23, 'artificial', 77),
5 (4, 'Sara', 19, 'computer', 88),
6 (5, 'Zain', 20, 'data', 87),
7 (6, 'Hassan', 26, 'electrical', 70),
8 (7, 'Areeba', 22, 'cyber', 88),
9 (8, 'Daniyal', 23, 'data', 81),
10 (9, 'Fatima', 20, 'cyber', 82);
```

Below the query editor, there are buttons for SELECT *, SELECT, INSERT, UPDATE, DELETE, Clear, Format, and a double arrow button. There is also a checkbox for "Bind parameters" and a "Get auto-saved query" button. At the bottom, there is a "Delimiter" dropdown set to semicolon, and checkboxes for "Show this query here again", "Retain query box", and "Rollback when finished". A "Go" button is at the bottom right.



Server: MySQL:3306 Database: dbs_lab4

Structure SQL Search Query Export Import Operations More

Current server: MySQL

Recent Favorites

New

dbs_lab4

New

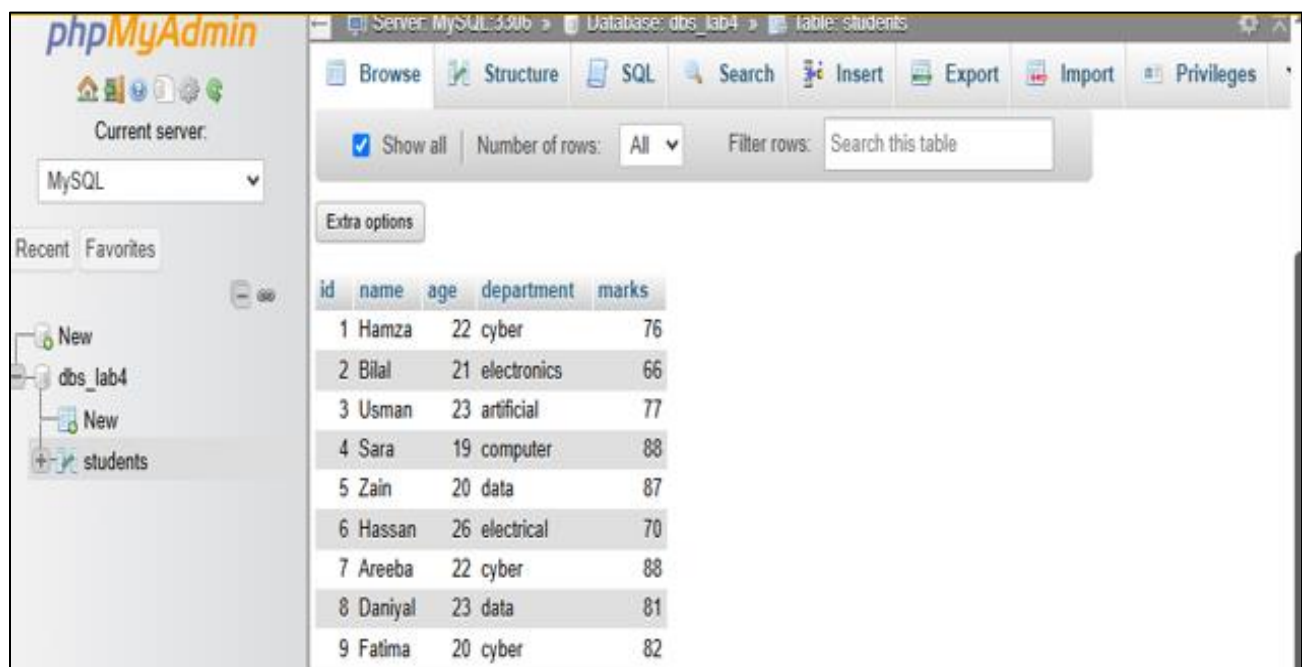
students

Show query box

✓ 9 rows inserted. (Query took 0.0057 seconds.)

```
INSERT INTO students (id, name, age, department, marks) VALUES (1, 'Hamza', 22, 'cyber', 76), (2, 'Bilal', 21, 'electronics', 66), (3, 'Usman', 23, 'artificial', 77), (4, 'Sara', 19, 'computer', 88), (5, 'Zain', 20, 'data', 87), (6, 'Hassan', 26, 'electrical', 70), (7, 'Areeba', 22, 'cyber', 88), (8, 'Daniyal', 23, 'data', 81), (9, 'Fatima', 20, 'cyber', 82);
```

[Edit inline][Edit][Create PHP code]



Server: MySQL:3306 Database: dbs_lab4 Table: students

Browse Structure SQL Search Insert Export Import Privileges

☒ Show all Number of rows: All Filter rows: Search this table

Extra options

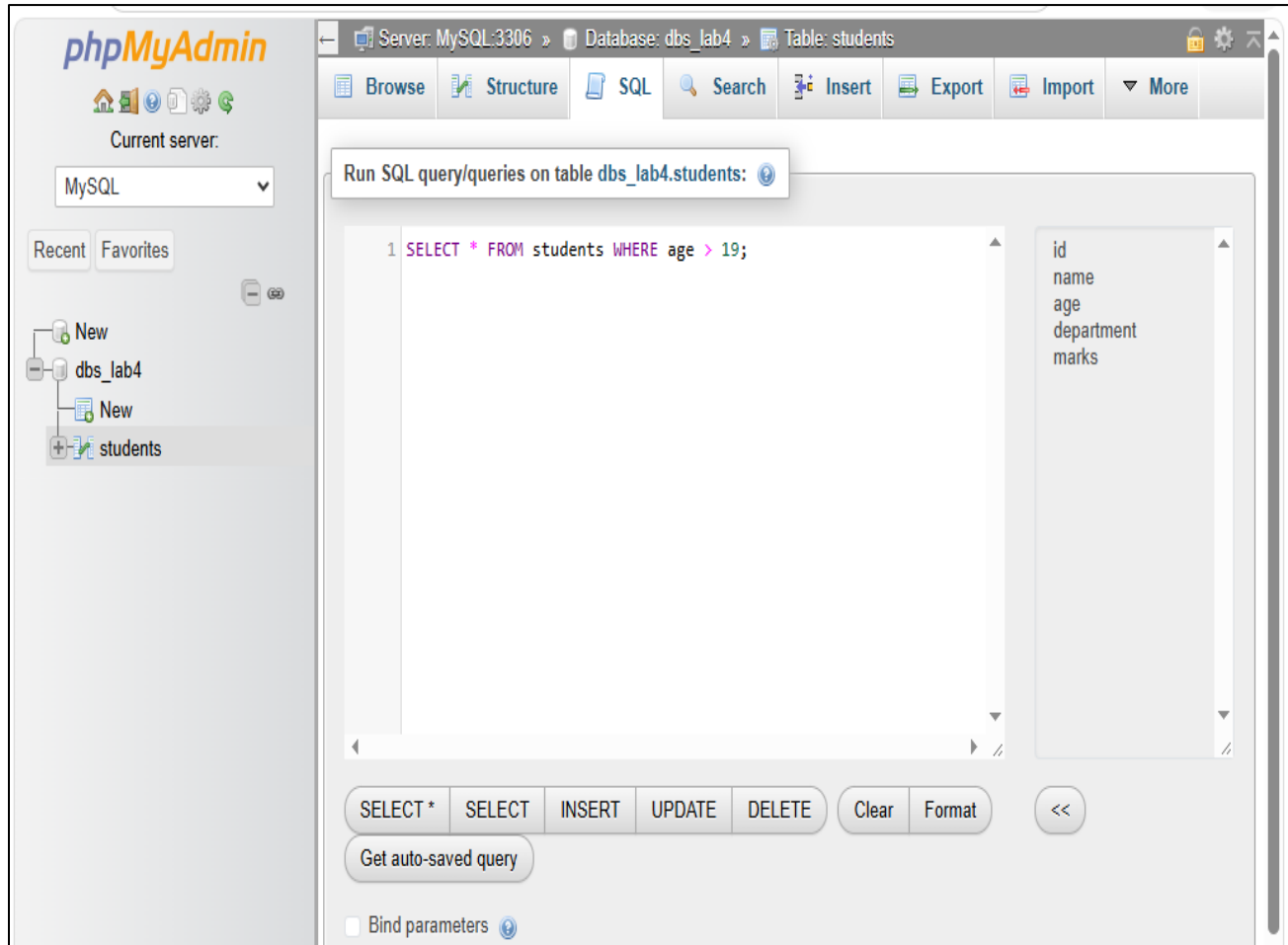
id	name	age	department	marks
1	Hamza	22	cyber	76
2	Bilal	21	electronics	66
3	Usman	23	artificial	77
4	Sara	19	computer	88
5	Zain	20	data	87
6	Hassan	26	electrical	70
7	Areeba	22	cyber	88
8	Daniyal	23	data	81
9	Fatima	20	cyber	82

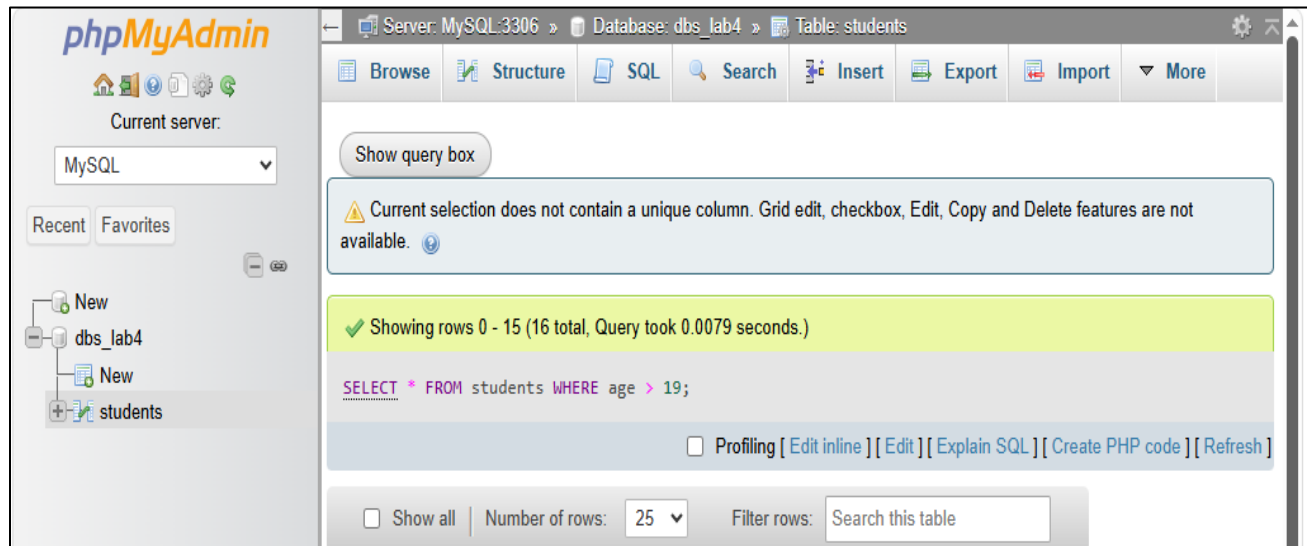
Output: Records inserted and displayed in the table

Step 3: SQL Clauses

The following queries were run to demonstrate each clause.

*Query 1 (WHERE): SELECT * FROM students WHERE age > 19;*





The screenshot shows the phpMyAdmin interface with the 'students' table selected. The SQL query executed is `SELECT * FROM students WHERE age > 19;`. The results show 16 rows, with the first 9 rows displayed in the table below.

Current server: MySQL

Recent Favorites

New
dbs_lab4
New
students

Server: MySQL:3306 » Database: dbs_lab4 » Table: students

Browse Structure SQL Search Insert Export Import More

Show query box

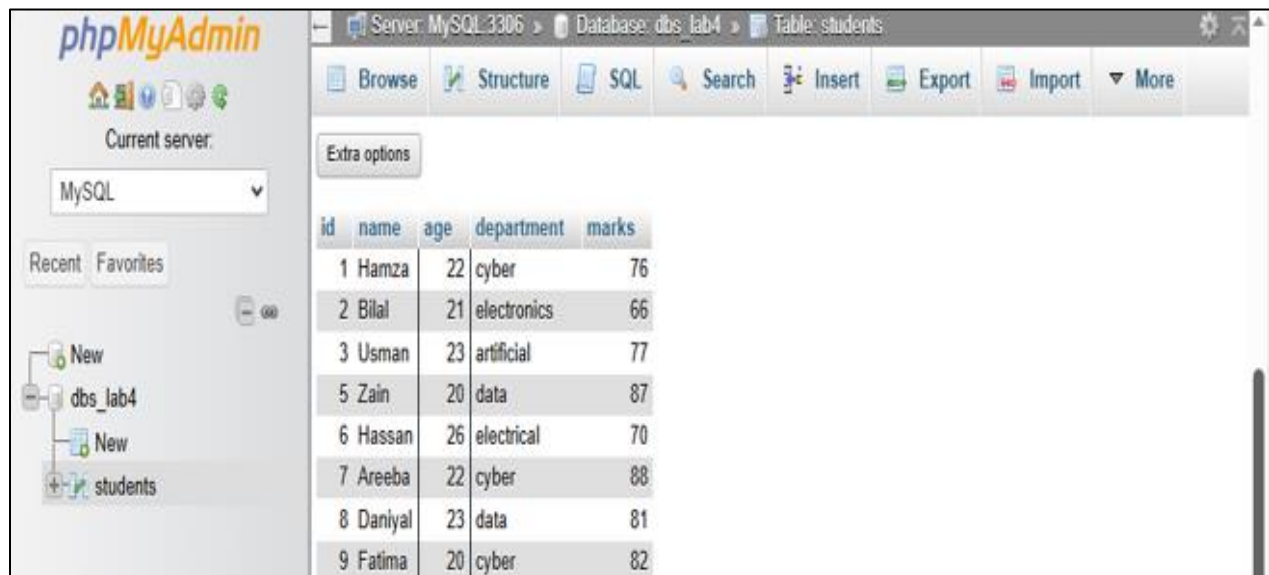
⚠ Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

✓ Showing rows 0 - 15 (16 total, Query took 0.0079 seconds.)

`SELECT * FROM students WHERE age > 19;`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table



The screenshot shows the phpMyAdmin interface with the 'students' table selected. The table structure and data are displayed below.

Current server: MySQL

Recent Favorites

New
dbs_lab4
New
students

Server: MySQL:3306 » Database: dbs_lab4 » Table: students

Browse Structure SQL Search Insert Export Import More

Extra options

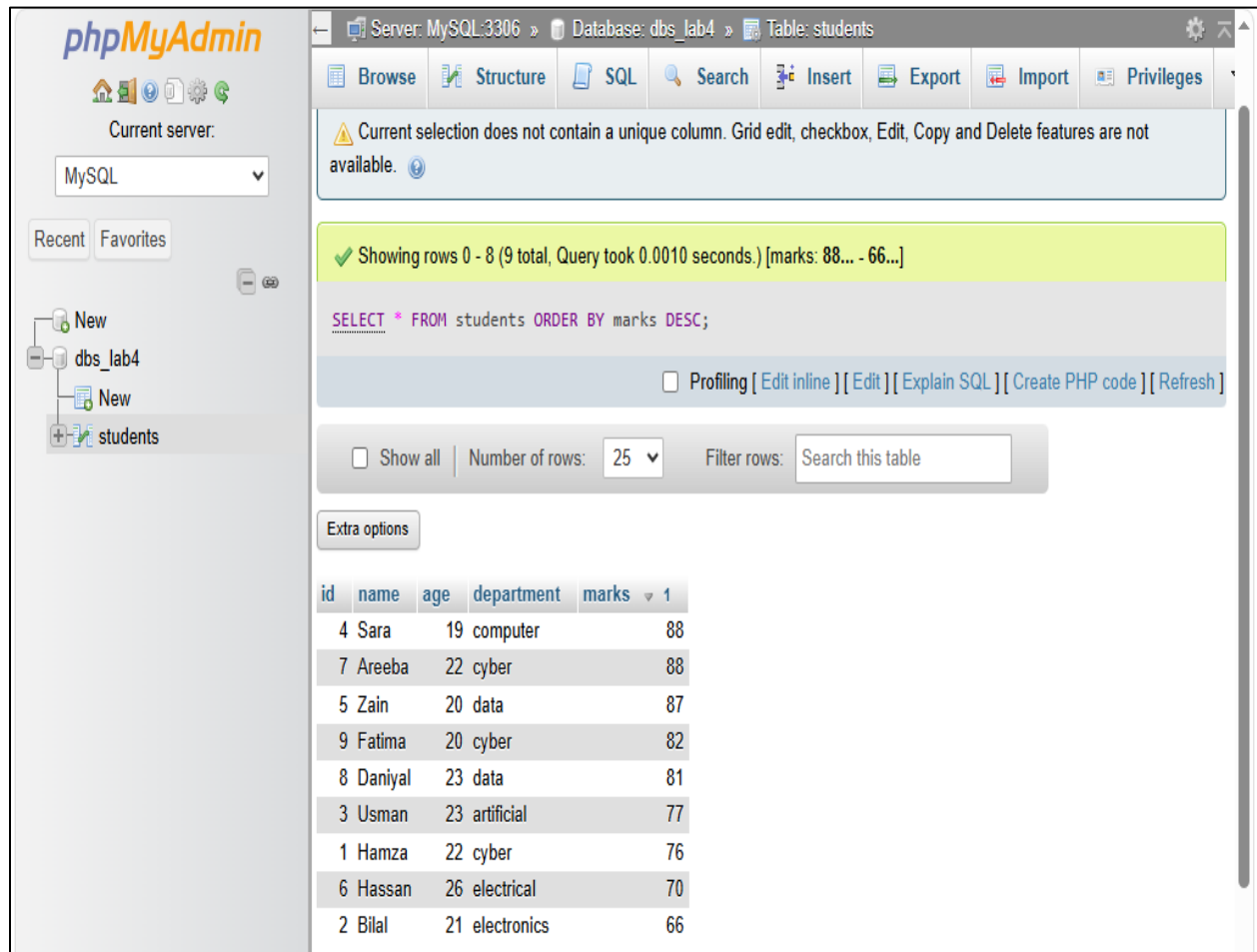
id	name	age	department	marks
1	Hamza	22	cyber	76
2	Bilal	21	electronics	66
3	Usman	23	artificial	77
5	Zain	20	data	87
6	Hassan	26	electrical	70
7	Areeba	22	cyber	88
8	Daniyal	23	data	81
9	Fatima	20	cyber	82

Output: Students older than 19

Query 2 (ORDER BY): *SELECT * FROM students ORDER BY marks DESC;*

The screenshot shows the phpMyAdmin interface with the SQL tab selected. The query editor contains the query: `1 SELECT * FROM students ORDER BY marks DESC;`. The table structure for 'students' is displayed on the right, showing columns: id, name, age, department, and marks. Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and '<<'. There is also a 'Get auto-saved query' button and a 'Bind parameters' checkbox. At the bottom, there is a 'Delimiter' dropdown set to semicolon, and checkboxes for 'Show this query here again', 'Retain query box', and 'Rollback when finished'. The 'Enable foreign key checks' checkbox is checked, and a 'Go' button is present.

The screenshot shows the phpMyAdmin interface with the SQL tab selected. The query editor contains the query: `SELECT * FROM students ORDER BY marks DESC;`. The results are displayed in a table with 18 rows. The first row is highlighted in green. The table structure for 'students' is displayed on the right, showing columns: id, name, age, department, and marks. Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and '<<'. There is also a 'Get auto-saved query' button and a 'Bind parameters' checkbox. At the bottom, there is a 'Delimiter' dropdown set to semicolon, and checkboxes for 'Show this query here again', 'Retain query box', and 'Rollback when finished'. The 'Enable foreign key checks' checkbox is checked, and a 'Go' button is present.



The screenshot shows the phpMyAdmin web interface. The top navigation bar indicates the current server is MySQL, the database is db4_lab4, and the selected table is students. A warning message states: "Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available." Below this, a green status bar shows "Showing rows 0 - 8 (9 total, Query took 0.0010 seconds.) [marks: 88... - 66...]". The SQL query editor contains the query: `SELECT * FROM students ORDER BY marks DESC;`. Below the query editor are links for Profiling, Edit inline, Edit, Explain SQL, Create PHP code, and Refresh. The table view shows 9 rows, with the first 8 visible. The columns are id, name, age, department, and marks. The rows are sorted by marks in descending order.

id	name	age	department	marks
4	Sara	19	computer	88
7	Areeba	22	cyber	88
5	Zain	20	data	87
9	Fatima	20	cyber	82
8	Daniyal	23	data	81
3	Usman	23	artificial	77
1	Hamza	22	cyber	76
6	Hassan	26	electrical	70
2	Bilal	21	electronics	66

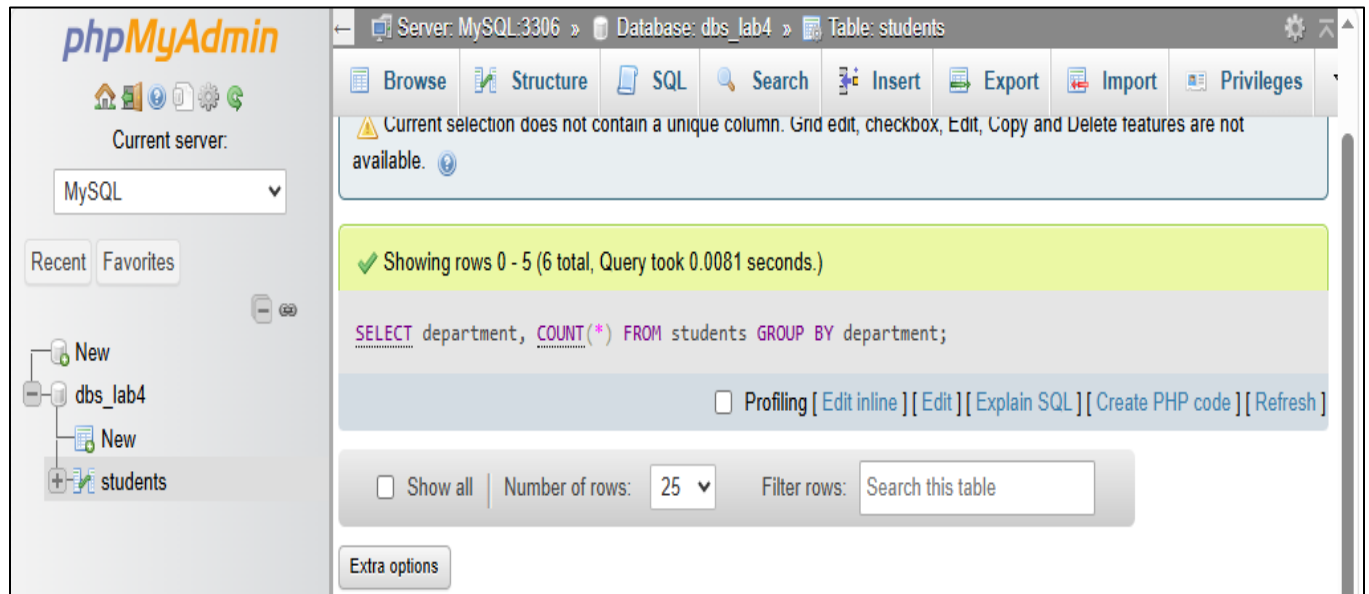
Output: Students sorted by marks (highest first)

Query 3 (GROUP BY): *SELECT department, COUNT(*) FROM students GROUP BY department;*

The screenshot shows the phpMyAdmin web interface. The top navigation bar indicates the current context: Server: MySQL:3306, Database: dbs_lab4, and Table: students. Below this, a toolbar contains buttons for Browse, Structure, SQL, Search, Insert, Export, Import, and More. The left sidebar shows the database structure with 'dbs_lab4' selected and 'students' as the current table. The main area is titled 'Run SQL query/queries on table dbs_lab4.students:' and contains a text input field with the following SQL query:

```
1 SELECT department, COUNT(*) FROM students GROUP BY department;
```

To the right of the query input is a preview window showing the table structure of 'students' with columns: id, name, age, department, and marks. Below the query input, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and '<<'. There is also a 'Get auto-saved query' button and a checkbox for 'Bind parameters'.



The screenshot shows the phpMyAdmin interface. The top navigation bar indicates the current selection is the 'students' table in the 'dbs_lab4' database. A warning message states: "Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available." Below this, a green status bar confirms: "Showing rows 0 - 5 (6 total, Query took 0.0081 seconds.)". The SQL query entered is: `SELECT department, COUNT(*) FROM students GROUP BY department;`. Below the query, there are links for "Profiling", "Edit inline", "Edit", "Explain SQL", "Create PHP code", and "Refresh". At the bottom of the query area, there are controls for "Show all", "Number of rows" (set to 25), and a "Filter rows" search box. An "Extra options" button is also present.

department	COUNT(*)
cyber	6
electronics	2
artificial	2
computer	2
data	4
electrical	2

Below the table, there are controls for "Show all", "Number of rows" (set to 25), and a "Filter rows" search box. At the bottom, a "Query results operations" section contains buttons for "Print", "Copy to clipboard", "Export", "Display chart", and "Create view".

Output: Number of students in each department

Query 4 (HAVING): *SELECT department, COUNT(*) FROM students GROUP BY department HAVING COUNT(*) > 1;*

The screenshot shows the phpMyAdmin interface with the 'students' table selected in the 'dbs_lab4' database. The SQL query editor is active, displaying the query: `1 SELECT department, COUNT(*) FROM students GROUP BY department HAVING COUNT(*) > 1;`. The table structure on the right lists columns: id, name, age, department, and marks. Below the query editor are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and '<<'. There is also a 'Get auto-saved query' button and a 'Bind parameters' checkbox.

The screenshot shows the phpMyAdmin interface displaying the results of the query. A warning message at the top states: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' Below this, a green banner indicates 'Showing rows 0 - 5 (6 total, Query took 0.0007 seconds.)'. The query text is repeated: `SELECT department, COUNT(*) FROM students GROUP BY department HAVING COUNT(*) > 1;`. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. At the bottom, there are controls for 'Show all', 'Number of rows' (set to 25), and a 'Filter rows' search box.

Extra options

department	COUNT(*)
cyber	6
electronics	2
artificial	2
computer	2
data	4
electrical	2

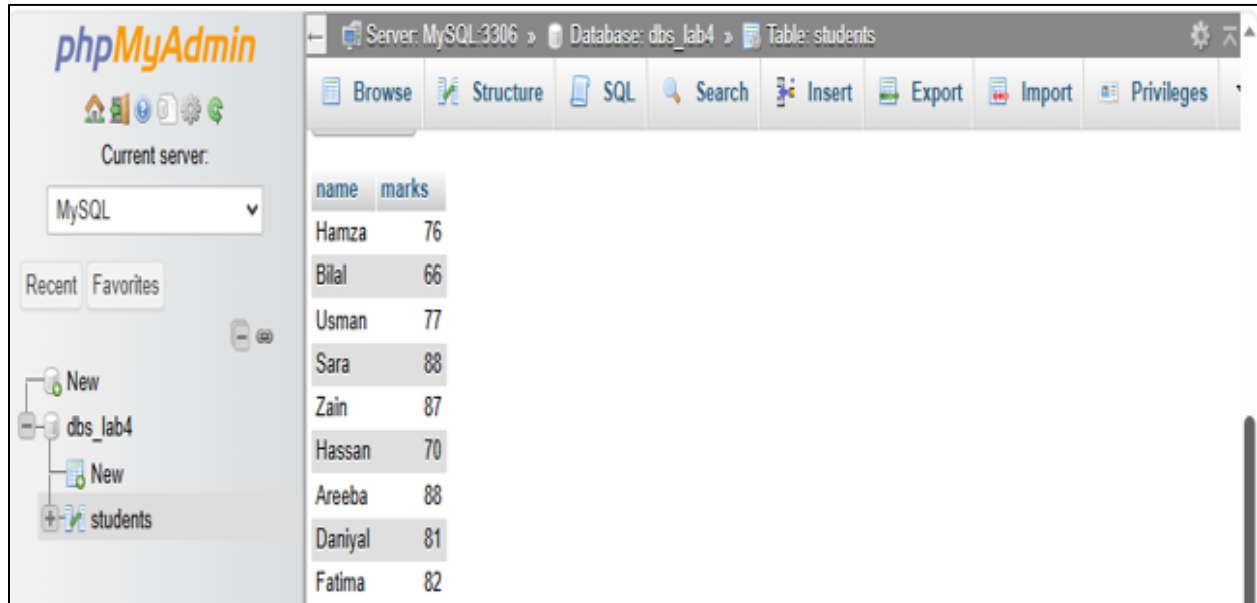
☐ Show all | Number of rows: 25 ▼ | Filter rows:

Output: Departments with more than one student

Query 5 (FROM): *SELECT name, marks FROM students;*

The screenshot shows the phpMyAdmin interface for the 'students' table in the 'dbs_lab4' database. The 'SQL' tab is selected, and the query editor contains the query: `1 SELECT name, marks FROM students;`. The table structure is visible on the right, showing columns: id, name, age, department, and marks. Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and '<<'. There is also a 'Get auto-saved query' button and a checkbox for 'Bind parameters'.

The screenshot shows the phpMyAdmin interface displaying the results of the query. A message at the top states: "Current selection does not contain a unique column. The edit, insertion, edit, copy and delete features are not available." Below this, a green bar indicates: "Showing rows 0 - 17 (18 total, Query took 0.0012 seconds.)". The query `SELECT name, marks FROM students;` is shown. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. At the bottom, there are options to 'Show all', 'Number of rows' (set to 25), and a 'Filter rows' search box. An 'Extra options' button is also present.



Server: MySQL:3306 > Database: db4_lab4 > Table: students

Current server: MySQL

Recent Favorites

New db4_lab4 New students

name	marks
Hamza	76
Bilal	66
Usman	77
Sara	88
Zain	87
Hassan	70
Areeba	88
Daniyal	81
Fatima	82

Output: Only the name and marks columns

Query 6 (LIMIT): *SELECT * FROM students LIMIT 3;*

The screenshot displays the phpMyAdmin web interface. The top navigation bar shows the current context: Server: MySQL3306, Database: dbs_lab4, and Table: students. Below this, a menu bar includes options like Browse, Structure, SQL, Search, Insert, Export, Import, and More. The left sidebar contains the phpMyAdmin logo, a 'Current server' dropdown set to 'MySQL', and a tree view of the database structure showing 'dbs_lab4' and its table 'students'. The main area is divided into two panes. The left pane contains a text editor with the SQL query: `1 SELECT * FROM students LIMIT 3;`. The right pane shows the table structure for 'students' with columns: id, name, age, department, and marks. At the bottom, there are buttons for query execution: 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and '<<'. There is also a 'Get auto-saved query' button and a 'Bind parameters' checkbox.

phpMyAdmin

Current server: MySQL

Recent Favorites

New
dbs_lab4
New
students

Server: MySQL3306 » Database: dbs_lab4 » Table: students

Browse Structure SQL Search Insert Export Import More

1 `SELECT * FROM students LIMIT 3;`

id
name
age
department
marks

SELECT * SELECT INSERT UPDATE DELETE Clear Format <<

Get auto-saved query

☐ Bind parameters

The screenshot shows the phpMyAdmin web interface. The top navigation bar indicates the current context: Server: MySQL3306 » Database: dbs_lab4 » Table: students. The left sidebar shows the database structure with 'dbs_lab4' selected and 'students' as the current table. The main panel displays a toolbar with options like Browse, Structure, SQL, Search, Insert, Export, Import, and Privileges. Below the toolbar, a message states: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' A green status bar indicates 'Showing rows 0 - 2 (3 total, Query took 0.0003 seconds.)'. The SQL query entered is 'SELECT * FROM students LIMIT 3;'. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A table titled 'Query results operations' shows the first three rows of the 'students' table:

id	name	age	department	marks
1	Hamza	22	cyber	76
2	Bilal	21	electronics	66
3	Usman	23	artificial	77

At the bottom, there are buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

Output: First 3 rows only

Query 7 (LIMIT with offset): *SELECT * FROM students LIMIT 0, 2;*

The screenshot shows the phpMyAdmin web interface. The top navigation bar indicates the current server is MySQL, the database is db4, and the selected table is students. The left sidebar shows the database structure with 'db4' selected and 'students' as the active table. The main area displays a SQL query editor with the query: `1 SELECT * FROM students LIMIT 0, 2;`. Below the editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and '<<'. A 'Get auto-saved query' button is also present. At the bottom, there is a checkbox for 'Bind parameters'.

id	name	age	department	marks
----	------	-----	------------	-------

The screenshot shows the phpMyAdmin web interface. The left sidebar displays the database structure with 'db Lab4' and 'students' table. The main panel shows the 'students' table with 2 rows. A query box contains the SQL: `SELECT * FROM students LIMIT 0, 2;`. The query results are displayed in a table with columns: id, name, age, department, marks.

id	name	age	department	marks
1	Hamza	22	cyber	76
2	Bilal	21	electronics	66

Output: 2 rows starting from offset 0

Query 8 (LIMIT with offset): *SELECT * FROM students LIMIT 2, 3;*

The screenshot shows the phpMyAdmin web interface. The top navigation bar indicates the current context: Server: MySQL:3306, Database: dbs_lab4, and Table: students. The left sidebar shows the database structure with 'dbs_lab4' selected and 'students' as the current table. The main area displays a SQL query editor with the query: `1 SELECT * FROM students LIMIT 2, 3;`. Below the query editor, there are buttons for 'SELECT *', 'SELECT', 'INSERT', 'UPDATE', 'DELETE', 'Clear', 'Format', and '<<'. A 'Get auto-saved query' button is also present. At the bottom, there is a checkbox for 'Bind parameters'. On the right side of the query editor, a list of table columns is visible: id, name, age, department, and marks.

The screenshot shows the phpMyAdmin web interface. The top navigation bar indicates the current context: Server: MySQL:3306, Database: dbs_lab4, and Table: students. A toolbar with icons for Browse, Structure, SQL, Search, Insert, Export, Import, and Privileges is visible. On the left sidebar, the 'Current server' is set to 'MySQL', and the database 'dbs_lab4' is selected, showing a table named 'students'. The main panel displays a 'Show query box' section with a warning message: 'Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.' Below this, a green status bar indicates 'Showing rows 2 - 4 (3 total, Query took 0.0005 seconds.)'. The SQL query entered is 'SELECT * FROM students LIMIT 2, 3;'. Below the query, there are links for 'Profiling', 'Edit inline', 'Edit', 'Explain SQL', 'Create PHP code', and 'Refresh'. A section titled 'Extra options' contains a table with the following data:

id	name	age	department	marks
3	Usman	23	artificial	77
4	Sara	19	computer	88
5	Zain	20	data	87

At the bottom, a 'Query results operations' section includes buttons for 'Print', 'Copy to clipboard', 'Export', 'Display chart', and 'Create view'.

Output: 3 rows after skipping the first 2

Results:

The students table was created and nine records were inserted successfully. The INSERT statement was executed twice, so each record appears two times in the table (18 rows in total). Each clause query was then executed and gave the expected output.

The WHERE clause returned 16 rows, because the records of Sara (age 19) were excluded by the condition age > 19. ORDER BY returned the students from the highest marks to the lowest, with Sara and Areeba (88 marks) at the top and Bilal (66 marks) at the bottom. GROUP BY divided the students into 6 departments: cyber (6), data (4), and electronics, artificial, computer and electrical (2 each). HAVING COUNT(*) > 1 returned all 6 departments, since each of them has more than one row. The FROM query displayed only the name and marks columns. LIMIT 3 returned the first three rows (Hamza, Bilal, Usman), LIMIT 0, 2 returned the first two rows (Hamza, Bilal), and LIMIT 2, 3 skipped the first two rows and returned the next three (Usman, Sara, Zain).

Conclusion:

SQL clauses control how the data returned by a query is filtered, ordered, grouped and limited. WHERE filters individual rows before grouping, while HAVING filters the groups after GROUP BY has been applied. ORDER BY arranges the results, FROM defines the source table and columns, and LIMIT restricts the number of rows returned. This lab provided hands-on practice with these clauses and helped build a clearer understanding of how data can be queried in a MySQL database using phpMyAdmin.